

Program Name: Bachelor of Computer Applications –Online

Semester: 1

1. Prerequisite: Basic approach of problem-solving methods

2. Rationale: The objective of this course is to familiarize students with concepts of fundamentals of information technology along with developing the logic for solving a given problem using the Procedure-Oriented Language C & Object-Oriented Language C++ for construction of code.

3. Course Content

Sr.	Topics	Weightage
1	Introduction to C Programming History, Algorithm and flowchart, Structure of C Elements of C: Character set, C Tokens, Keywords Identifiers, Variables, Constant Data Types, Comments, C Programming Applications and Importance, Operators: What is operator? Types of operators,	10 %
2	Pre-processors, Storage Classes Storage Classes: Automatic, External, Static and Register Variables, Decision Making / Control Statements: If, If Else, Nested if, Switch, looping statements: For, nested for, While, do while, other statements: Break, Continue, Goto, exit.	10 %
3	Array Declaration, Initialization, Access of one dimensional & two-dimensional arrays, Programs using one- and two-dimensional arrays: Adding multiplying, transposing matrices: sorting and searching array	10 %
4	Function, Structure and Union Definition, need of function, Types of function, Built-in and User define Functions, User define Functions, Categories of functions: With/without arguments, With/without return values, Recursion, Functions with arrays, The scope, visibility & lifetime of variables. Structure definition, Giving values to members, Structure initialization, Comparison of structure variables, Arrays of structures, Arrays within structures, Structures within structures, Structures & functions, Unions Size of structures	20 %
5	Pointer and Working with Strings Understanding pointers, Accessing the address of a variable, Declaring & initializing pointers, accessing a variable through its pointer, Pointer expression, Pointer increments & scale factor, Pointers & arrays, Passing pointer variables as function arguments. Declaring & initializing string variables, reading strings from terminal, writing strings to screen, Arithmetic operations on characters, putting strings together, comparison of two strings, string handling functions, table of strings.	20 %
6	File Management in C Introduction to file management and its functions	4 %

7	Introduction to C++ Programming Procedure Oriented Programming, Object Oriented Programming, Purpose of object-oriented Programming, Procedural Vs Object Oriented Programming, and Principles of object-oriented programming. Benefits and Applications of OOP. Overview of C++, Program structure, namespace, identifiers, Data types in C++, variables, constants, operators, typecasting, control structures.	16 %
8	Functions in C++ Basics of functions, Parameter Passing Mechanism, Inline function, Macro, Macro Vs Inline function, Function Overloading, Default Arguments.	10 %

4. Reference Books

1. **Programming in ANSI C**
By E. Balaguruswamy | Tata McGraw-Hill
2. **The C Programming Language**
By Brian W. Kerningham and Dennis M. Ritchie | PHI
3. **Programming with C**
By K.R. Venugopal and Sudeep R Prasad | Tata McGraw-Hill Education
4. **Let Us C**
By Yeshavant Kanetkar | BPB Publications
5. **Object Oriented Programming with C++ - 5th edition**
By E. Balagurusamy | Tata McGraw Hill | Fifth Edition
6. **The Complete Reference C++,**
By by Herbert Schildt | TMH

5. Course Outcome

After Learning the Course, the students shall be able to:

1. Interpret syntax and semantics of C programming language.
2. Develop, execute and debug programs.
3. Develop the algorithms for solving Engineering problems.
4. Develop confidence for self-education and ability for life-long learning needed for Computer language.
5. Recognize features of object-oriented design such as encapsulation, polymorphism, inheritance, and composition of systems based on object identity.

6. List of Practical

1. **WAP to Add Two Integers**
2. **WAP to Swap Two Numbers**
3. **WAP to Find Area of Circle**

4. **WAP to Find a Number is Even or Odd**
5. **WAP to Find Factorial of a Number**
6. **WAP to print Fibonacci Series**
7. **Write a menu-driven program using Switch case to create calculator**
8. **WAP to swap two numbers using function**
9. **WAP to swap two integers using pointers**
10. **WAP to pass string to a function and find length of it**
11. **WAP to concatenate two strings and copy the string 1 to string 2**
12. **Write a program in C++ demonstrating the public, protected and private parameters.**
13. **Write a program in C++ to exchange the content of two variables using call by reference**